

Renormalization Group Flow as Dissipative Gradient Descent: A Thermodynamic Basis for Model Selection

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The Renormalization Group (RG) describes how physical systems change appearance across scales, governed by the flow of coupling constants (beta functions). In this paper, we propose that RG flow can be formalized as a gradient descent process on a Variational Free Energy landscape defined over the manifold of model parameters. We demonstrate that the irreversible flow toward infrared fixed points is isomorphic to a thermodynamic relaxation process. By invoking Landauer’s Principle, we argue that the “coarse-graining” of degrees of freedom—effectively the erasure of microscopic information—generates a lower bound on heat dissipation. This framework, which we term “Reflexive Renormalization,” provides a thermodynamic arrow of time emergent from the minimization of model complexity.

I. INTRODUCTION

In Quantum Field Theory (QFT) and Statistical Mechanics, the Renormalization Group (RG) flow describes the evolution of a physical system’s effective action as the observation scale changes [1]. Typically, this flow is treated as a mathematical consequence of scaling transformations. However, recent work in Information Geometry suggests that RG flow follows geodesics or gradient flows on a statistical manifold [2, 3].

Simultaneously, the thermodynamics of computation [4, 5] establishes a physical cost for information processing. Specifically, Landauer’s Principle dictates that the erasure of information (a logically irreversible operation) must be accompanied by the dissipation of heat into the environment.

This paper attempts to unify these perspectives. We posit that the physical universe behaves as a self-evidencing system minimizing a Variational Free Energy functional $\mathcal{F}$. We identify the RG flow of coupling constants $\frac{d\mathbf{g}}{d\tau}$ not merely as a change in scale, but as a dynamical optimization process—a “Reflexion”—that prunes non-parsimonious parameters. We derive the thermodynamic cost of this pruning, suggesting that the arrow of time is driven by the metabolic cost of model simplification.

II. THE STATISTICAL MANIFOLD OF THEORIES

Consider a physical theory described by a Hamiltonian $\mathcal{H}(\phi; \mathbf{g})$, where ϕ represents the fields and $\mathbf{g} = \{g_1, g_2, \dots, g_N\}$ is a set of coupling constants parameterizing the theory (e.g., mass, interaction strength).

This defines a family of probability distributions

(Gibbs states) over the configuration space:

$$P_{\mathbf{g}}(\phi) = \frac{1}{Z(\mathbf{g})} e^{-\mathcal{H}(\phi; \mathbf{g})} \quad (1)$$

where $Z(\mathbf{g})$ is the partition function.

The space of all such theories forms a Riemannian manifold $\mathcal{M}$ equipped with the Fisher Information Metric $\mathcal{G}_{ab}$, which measures the distinguishability of neighboring theories:

$$\mathcal{G}_{ab}(\mathbf{g}) = \left\langle \frac{\partial \ln P_{\mathbf{g}}}{\partial g_a} \frac{\partial \ln P_{\mathbf{g}}}{\partial g_b} \right\rangle \quad (2)$$

III. RG FLOW AS GRADIENT DESCENT

Standard RG analysis defines the beta functions $\beta_a(\mathbf{g})$ as the rate of change of couplings with respect to the logarithmic scale $\tau = \ln(\mu/\Lambda)$:

$$\frac{dg_a}{d\tau} = \beta_a(\mathbf{g}) \quad (3)$$

We propose that this flow is not arbitrary but follows the path of steepest descent on a free energy potential $\Phi(\mathbf{g})$. This is known as the gradient flow equation [6]:

$$\frac{dg_a}{d\tau} = -\mathcal{G}^{ab} \frac{\partial \Phi}{\partial g_b} \quad (4)$$

where $\mathcal{G}^{ab}$ is the inverse Fisher metric, acting as the “mobility” tensor for the flow.

A. The Variational Free Energy Functional

We identify the potential $\Phi(\mathbf{g})$ with the **Variational Free Energy** (or Relative Entropy) between the effective theory at scale τ and the “true” microscopic distribution P_{UV} :

$$\Phi(\mathbf{g}) = D_{KL}[P_{\mathbf{g}} || P_{UV}] + \lambda C(\mathbf{g}) \quad (5)$$

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Here, the first term enforces fidelity to the data (accuracy), while the second term $C(\mathbf{g})$ is a complexity penalty (parsimony) regularized by λ .

The RG flow, therefore, drives the system toward a configuration that maximizes parsimony while retaining relevant information about the UV state. We term this operator "Reflexion" (R), as the system recursively updates its own parameters to minimize Φ .

IV. THE LANDAUER BRIDGE

We now consider the thermodynamic implications of Eq. (4). The rate of change of the potential Φ along the flow is:

$$\frac{d\Phi}{d\tau} = \frac{\partial\Phi}{\partial g_a} \frac{dg_a}{d\tau} = -\mathcal{G}^{ab} \frac{\partial\Phi}{\partial g_a} \frac{\partial\Phi}{\partial g_b} \quad (6)$$

Since the Fisher metric $\mathcal{G}$ is positive definite, $\frac{d\Phi}{d\tau} \leq 0$. This is an H-theorem for the renormalization group: the free energy monotonically decreases along the flow.

A. Information Erasure and Heat

As the flow progresses toward the Infrared (IR) fixed point, irrelevant couplings g_{irr} decay to zero. This constitutes a reduction in the effective degrees of freedom of the model.

Let ΔI be the reduction in Shannon information (entropy difference) between the UV and IR states:

$$\Delta I = S[P_{IR}] - S[P_{UV}] < 0 \quad (7)$$

According to Landauer's Principle, the erasure of information ΔI in a physical system coupled to a thermal

bath at temperature T must dissipate heat ΔQ :

$$\Delta Q \geq -k_B T \Delta I \quad (8)$$

Substituting the trajectory of the parameters from the gradient flow:

$$\frac{dQ_{diss}}{d\tau} \approx k_B T \cdot \text{Tr}(\mathcal{G}_{ab} \beta^a \beta^b) \quad (9)$$

This result implies that the "Reflexion" of the system—the process of simplifying the theory by flowing to the IR—is inherently dissipative. The thermodynamic arrow of time aligns with the direction of parameter reduction.

V. DISCUSSION

We have derived a thermodynamic basis for Renormalization Group flow. By treating the space of coupling constants as a statistical manifold, we identified the beta functions as gradient vectors minimizing a variational free energy.

This formalism supports the "PCIR" hypothesis (Parsimony, Context, Information, Reflexion), specifically validating the **Reflexion (R)** component. It suggests that: 1. Physical laws evolve toward parsimony (IR fixed points). 2. This evolution is driven by a minimization of information divergence. 3. The process generates heat, linking the abstract flow of model parameters to physical entropy production.

Future work will explore the coupling of this thermodynamic flow to the emergent geometry of the statistical manifold, potentially linking dissipative RG flow to the expansion of spacetime.

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